Information Coefficient and Information Ratio

Measuring Forecasting Skill and Risk-Adjusted Active Return

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 4

1 Information Coefficient (IC)

The information coefficient measures the predictive power of a forecasting model as the correlation between forecast and realized returns:

$$\text{IC} = \text{Corr}\big(\text{forecast}_i, \text{realized return}_i\big)$$

The correlation is typically computed cross-sectionally across all assets in a universe at each time period, then averaged over time.

1.1 Interpretation and typical values

The IC ranges from -1 to $+1$:

- $\text{IC} > 0$: positive predictive skill (forecasts align with realized returns);
- $\text{IC} = 0$: no predictive skill;
- $\text{IC} < 0$: negative predictive skill (can be exploited by reversing the signal).

$\lvert\text{IC}\rvert$	**Forecasting ability**
> 0.10	Excellent (rare)
0.05–0.10	Strong
0.02–0.05	Moderate
< 0.02	Weak or negligible

Even modest IC values can be valuable when applied across many securities over time, as the Fundamental Law of Active Management (Section 4) makes precise.

1.2 Statistical significance

To be statistically significant, an IC must be large relative to its standard error. The t-statistic for $H_0\colon \text{IC} = 0$ is

$$t = \frac{\text{IC}\sqrt{N}}{\sqrt{1-\text{IC}^2}} \approx \text{IC}\sqrt{N} \quad \text{for small IC},$$

where N is the number of observations (time periods or asset–period pairs).

2 Information Ratio (IR)

The information ratio measures the risk-adjusted active return of a portfolio relative to a benchmark: the excess return generated per unit of active risk taken.

$$\text{IR} = \frac{\alpha}{\omega} = \frac{E[R_p - R_b]}{\text{Std}(R_p - R_b)}$$

where

- R_p — portfolio return; R_b — benchmark return;
- $\alpha = E[R_p - R_b]$ — expected active return;
- $\omega = \text{Std}(R_p - R_b)$ — tracking error (active risk).

Higher IR means better risk-adjusted active performance: $\text{IR} > 0.5$ is typically considered good and $\text{IR} > 1.0$ excellent. IR measures the consistency of outperformance relative to the variation in active returns.

3 Information Ratio vs. Sharpe Ratio

Both ratios measure risk-adjusted return, but against different references:

$$\text{SR} = \frac{E[R_p - r_f]}{\text{Std}(R_p)}, \qquad \text{IR} = \frac{E[R_p - R_b]}{\text{Std}(R_p - R_b)}.$$

Aspect	Sharpe ratio	Information ratio
Reference	Risk-free rate	Custom benchmark (e.g. an index)
Numerator	Total excess return over r_f	Active return over the benchmark
Risk measure	Total volatility	Tracking error (active risk)
Purpose	Absolute performance	Relative performance
Use case	Choosing among stand-alone portfolios	Evaluating active managers

If the benchmark is a constant risk-free rate, the information ratio equals the Sharpe ratio; the IR generalizes the idea to any benchmark.

4 Fundamental Relationship: IC and IR

The **Fundamental Law of Active Management** (Grinold and Kahn) links forecasting skill to realized performance:

$$\text{IR} \approx \text{IC} \times \sqrt{\text{BR}}$$

where BR (breadth) is the number of independent investment decisions per year. Value therefore has two sources:

1. **Skill (IC)**: the quality of the forecasts;
2. **Breadth (BR)**: the number of independent opportunities to apply that skill.

Key insights.

- Modest skill can generate substantial value when applied broadly: IC $= 0.05$ with BR $= 400$ gives IR $= 0.05 \times \sqrt{400} = 1.0$.
- IR can be raised by increasing IC (better forecasts) or BR (more independent bets).
- High-frequency strategies exploit breadth: many decisions with modest IC. Concentrated portfolios need a high IC to make up for low breadth.
- Multi-strategy approaches increase effective breadth; correlation among signals reduces it.

The law assumes optimal, risk-aware portfolio construction, no transaction costs or constraints, and independent decisions. In practice, constraints, costs, and signal correlation keep the realized IR below this theoretical value.